SEAT NO. CT-23025

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR (COMPUTER SCIENCE & INFORMATION TECHNOLOGY/ CS & IT (SPECIALIZATION IN "ARTIFICIAL INTELLIGENCE")
SPRING SEMESTER EXAMINATIONS 2025
Batch 2023

Time: 3 Hours | **Dated:11-06-2025**
Max.Marks:60

Computer Architecture & Organization – CS-252

CLO – 2: Bloom's Taxonomy Understanding (Cognitive Level C3) (PLO-3)

Q 1. Explain the Following Statements **10 Marks**

a) Why data removal from the cache not speed up the processor
b) The instruction DEC N inform the assembler about what?
c) Why we need Assembly Language when we have HLL and LLL?
d) What is Machine Code?
e) How to convert Assembly language to Machine Code.
f) What is an object file
g) Why compressed instruction consider to be faster in comparison to 64-bit instructions?
h) Why we use adder
i) Instruction having MOV and Add are called
j) The last statement of the source program is

CLO - 1 : Bloom's Taxonomy (Cognitive Level C2) (PLO-1)

Q2. (a) Use assembly language to make a program which swap two 8 bit numbers using third register **05 Marks**

CLO - 1: Bloom's Taxonomy (Cognitive Level C2)(PLO-1)

Q2. (b) Explain with the help of diagram Computer Top Level View **05 Marks**

CLO - 2 : Bloom's Taxonomy (Cognitive Level C3) (PLO-3)

Q3. (a) Interpret the Data path concepts diagram to answer the following questions. Show working where necessary **05 Marks**

a) What will be the size of ALUop bits if ALU has 38 Operation to perform
b) Is it necessary to have same size of Instruction Memory and Data Memory at the same time
c) If we have instruction of 32 bits and our Instruction memory is byte addressable. What will be the value of Adder we use to add with PC
d) If we have instruction of 32 bits and our Instruction memory is word addressable. What will be the value of Adder we use to add with PC
e) If we have instruction of 32 bits and our Instruction memory is double word addressable. What will be the value of Adder we use to add with PC

CLO - 2 : Bloom's Taxonomy (Cognitive Level C3) (PLO-3)

Q3. (b) Interpret the following **05 Marks**

i. Computer Architecture
ii. I/OBR
iii. User Visible Register
iv. L1 Cache
v. Hit Ratio

P.T.O

CLO - 2: Bloom's Taxonomy (Cognitive Level C3) (PLO-3)
Q4. (a) Draw and explain processing of Cache Mapping **04 Marks**

CLO - 2: Bloom's Taxonomy (Cognitive Level C3) (PLO-3)
Q4. (b) Define and **interpret** General Purpose Registers **04 Marks**

CLO - 2: Bloom's Taxonomy (Cognitive Level C3) (PLO-3)
Q4. (c) By using the following information calculate the missing data
i. Main Memory 16B; Frame Size 4B (No of Frames =?) **02 Marks**
ii . Process Size 32B; Page Size 8B (No of Pages =?)

CLO - 1 : Bloom's Taxonomy (Cognitive Level C2) (PLO-1)
Q5. (a) Explain the Single Cycle Diagram in different stages. Cover each Block **05 Marks**
Insb

CLO - 1: Bloom's Taxonomy (Cognitive Level C2) (PLO-1)
Q5. (b) Explain LRU algorithm in detail **02 Marks**

CLO - 1: Bloom's Taxonomy (Cognitive Level C2) (PLO-1)
Q5. (c) Apply LRU Algorithm to the following data to calculate Page Fault (page frame = 4) **03 Marks**
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2
1, 9, 2, 5, 0, 1, 3, 1, 3, 2, 8, 3, 2, 3, 2, 5

CLO - 2: Bloom's Taxonomy (Cognitive Level C3) (PLO-3)
Q6. (a) What is DMA Controller? **Draw** the diagram to demonstrate the idea **05 Marks**

CLO - 2: Bloom's Taxonomy (Cognitive Level C3) (PLO-3)
Q6. (b) Demonstrate with the help of Waveforms of the following Assembly Language where necessary and give reason where it cannot be formed **05 Marks**

ADD AL,BL
MOV CL,3
MUL DL,AL
MOV BL,0
MOV CH,3
DIV BL,AL
MOV DL,120
MOV AL,2
JE AL,BL
JGE CL,BL
	JE AL,BL
	SUB AL,BL
	MOV CL,3
	DIV DL,AL
	MOV BL,0
	MOV CH,3
	DIV BL,AL
	MOV DL,120
	MOV AL,2
	JGE CL,BL
		DIV BL,AL
		MOV DL,120
		MOV AL,2
		JE AL,BL
			ADD AL,BL
			MOV CL,A
			MUL DL,AL
			MOV BL,0
			MOV CH,-05
			DIV BL,AL
			MOV DL,120
			MOV AL,2
			JE AL,BL
			JGE CL,BL
				MOV CL,3
				MUL DL,AL
				MOV BL,0
				MOV CH,3
				DIV BL,AL
				MOV DL,120
				MOV AL,2
				JE AL,BL
				JGE CL,BL
				JZ BL